

Predicting the Propensity to Alternans from Spatial Differences in the Restitution Portrait

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ABSTRACT: Introduction: Traditionally, it was believed that cardiac rhythm stability is governed by the slope of the restitution curve (RC), which relates the duration of an action potential to the preceding diastolic interval. However, a single RC does not exist: short-term memory causes rate-dependence, which leads to multiple distinct RCs. The restitution portrait (RP) permits simultaneous visualization of transient responses together with three different RCs: S1-S2 (SRC), constant-BCL (BRC), and dynamic (DRC). Here, we investigate spatial differences in the RP and correlate these differences with the tissue's propensity to develop alternans.

Methods and Results: Action potentials were measured in paced superfused bullfrog ventricle with two microelectrodes located ~1-2 mm from each other and ~1 mm from the pacing site. RPs were constructed for both locations and characterized quantitatively by the steady-state action potential durations (APDs) and the slopes of the three RCs. Spatial differences in these characteristics were analyzed as a function of pacing rate. The results show that spatial differences in APD and in SRC slope predict the tissue's propensity to alternans at all pacing rates; the differences in BRC and DRC slopes do so only at some pacing rates.

Conclusions: These results suggest a new diagnostic approach to identifying patients with vulnerability to arrhythmias, which involves pacing at slow rates and analyzing spatial differences in restitution properties. They also suggest that pharmacological treatments that decrease spatial differences in restitution properties may be effective in preventing arrhythmias.

Keywords: cardiac electrophysiology, APD alternans, restitution, restitution portrait, spatial gradients

Introduction

Ventricular fibrillation is one of the leading causes of death in the United States [1]. Thus there is great interest in finding a way to identify patients who are susceptible to fibrillation. A significant body of research concentrates on action potential duration (APD) alternans, a long-short alternation in APD that was found to be a precursor to more complex arrhythmias [2-5]. Researchers are interested in understanding the mechanisms underlying the transition from a stable 1:1 rhythm to alternans. A better understanding of this transition may lead to new screening procedures that will determine a patient's vulnerability to arrhythmias and to new treatments that will prevent patients from developing arrhythmias, rather than treat the arrhythmias after they occur.

An early attempt at predicting alternans in cardiac cells was by Nolasco and Dahlen [6]. They assumed that APD is a function of *only* the previous diastolic interval (DI), expressed mathematically as a mapping $APD_{n+1}=f(DI_n)$, where n is the beat number and f is a function defining what is referred to as the restitution curve. They argued that the rhythm stability was determined by the slope of the restitution curve: the rhythm is stable (1:1 stimulus:response pattern) when the slope is below one, and it transitions to alternans (2:2 pattern) when the slope becomes greater than one. This criterion is known as the restitution hypothesis and has been verified by some experimental [7,8] and computational [9,10] studies. However, more recent experimental studies have contradicted this hypothesis by showing that 2:2 responses occur with restitution slopes < 1 [11-13] and also that 1:1 responses occur with restitution slope > 1 [13-15]. One reason for the failure of the restitution hypothesis is short-term cardiac memory [16,17]. It

manifests itself as *rate-dependence*, where the restitution curve depends on the pacing protocol used to measure it [18].

The restitution portrait (RP) was recently developed as a tool to deal with rate-dependence and to help visualize the effects of memory on cardiac dynamics [19]. The RP combines steady-state responses (dynamic restitution curve, DRC) [8], responses to an extrastimulus (several segments of S1S2 restitution curves, SRC) [20], recovery beats after an extrastimulus (segments of the constant-BCL restitution curve, BRC) [14], and transient responses after a change in pacing rate. Recent studies have shown that a model with cardiac memory of the form $APD_{n+1} = g(DI_n, APD_n, DI_{n-1})$ can reproduce experimentally measured RPs both qualitatively and quantitatively [21]. Analysis of the stability of this model has led to a new criterion for alternans, in which a *combination* of slopes of SRC, BRC, and DRC has to exceed one [16]. However, this ‘memory’ stability criterion still fails when tested against experimental data [19].

A possible reason for the failure of the existing stability criteria is that they are derived from models of local, single-cell dynamics, but experimental measurements on which they are tested have been made in multicellular preparations. Previous experiments have shown spatial variation in several restitution properties, such as APD [4,11,15,22-29], slope of the DRC [22,26], and slope of the SRC [23,24,27-31]. As shown by some of these studies, spatial variation in restitution properties may be linked to the onset of arrhythmias [4,23-32]. Pastore *et al.* [4] found that spatial gradients of APD larger than 3 ms/mm were associated with conduction block and eventual fibrillation in guinea pig hearts. Aiba *et al.* [24] determined that areas of steep APD gradients in dog heart muscle were the point of origin of arrhythmias when they arose, but that a

steep APD gradient alone was not enough to trigger an arrhythmia. The use of APD gradients for predicting conduction block was further brought into question by Qin *et al.* [27], who found that steep APD gradients were not correlated with conduction block. Finally, Pak *et al.* [28] found that patients with inducible ventricular tachycardia had larger spatial APD gradients and larger spatial gradients of the SRC slope than a control group. These mixed results suggest that there is still much uncertainty about the effect of spatial heterogeneity of restitution properties on the stability of heart rhythm.

In this paper, we investigate spatial differences in restitution properties by performing simultaneous measurement of the RPs at two spatial locations in a bullfrog ventricle *in vitro*. The measurements are performed with microelectrodes, instead of optical mapping, to avoid the use of electro-mechanical uncouplers [34] and phototoxic effects of voltage sensitive dyes. The use of the RP technique permits the measurement of many restitution properties in a single experiment and provides a more comprehensive picture of the spatial differences of restitution properties than previous experiments. The bullfrog ventricle provides a homogeneous substrate [35-37] that permits us to assume that spatial differences are induced solely by the dynamics of paced cardiac tissue. Our study intends to answer two questions: (a) Are there dynamical, pacing-induced spatial differences in any of the restitution properties measured by the RP, such as the steady-state APD or slopes of the BRC, SRC, and DRC? (b) Are spatial differences in any of these properties predictive of the tissue's propensity to alternans?

Methods

Tissue Preparation

This study was performed in accordance with a protocol that conforms to the Research Animal Use Guidelines of the American Heart Association and was approved by the Duke University Institutional Animal Care and Use Committee. Seven bullfrogs, *Rana Catesbeiana*, were anesthetized and double-pithed. The heart was excised and the anterior surface of the ventricle was removed and pinned to the bottom of a perfusion chamber. The tissue was superfused with a standard Ringer's solution (100 mM NaCl, 2.70 mM KCl, 5.6 mM glucose, 1 mM hepes, 2.8 mM Na_2HPO_4 , 25 mM NaHCO_3 , 1.5 mM MgCl_2 , 1.80 mM CaCl_2 , buffered by CO_2) at room temperature. The tissue was paced at twice diastolic threshold with a silver bipolar electrode (Fig. 1A) at a basic cycle length (BCL) of 1,000 ms for 20 minutes before the data collection began.

Pacing Protocol

The tissue was paced using a perturbed downsweep protocol [19] that allowed us to collect all the data needed to construct the RP. A pacing sequence for one BCL is shown in Fig. 1B. Beginning at a long BCL (typically 1000 ms), the tissue is paced for 60 s (small dots) until steady state is achieved. Five steady-state paces (diamonds) are applied at the initial BCL before an S2 pace at BCL+50 ms ('+') is applied. This is followed by five recovery paces (filled circles) at the initial BCL and another S2 pace at BCL-50 ms ('x'). The sequence ends with five recovery paces (filled circles) at the initial BCL. The BCL is then decremented by 50 or 100 ms and the sequence of Fig. 1 is repeated. Pacing at decreasing BCLs continues until either a 2:1 or 2:2 response is seen. A typical trial consisted of 15 BCLs and lasted 20 minutes. Several trials were

performed on each animal. Only trials where both electrodes remained impaled for the majority of the BCLs were analyzed. Results from 19 trials in 7 animals are presented in this paper.

Seven trials in 4 animals showed steady-state alternans in at least one electrode, which we denote as ‘ALT’ trials; the remaining trials that did not show alternans are denoted as ‘noALT’ trials. Specifically, the final 4 steady-state responses (diamonds in Fig. 1B) were used to determine whether a trial was classified as ALT or noALT. For each BCL, we determined $\delta A = APD_{n+1} - APD_n$ where $n = 1 \dots 4$ are the steady-state beats. A response pattern was classified as displaying steady-state alternans if δA alternated in sign from beat to beat and $|\delta A| > 2$ ms (the error in APD measurement).

Electrical Recordings

Electrical signals were recorded simultaneously from two locations in the tissue using glass microelectrodes filled with 3 M KCl. Microelectrodes were placed 1-2 mm apart perpendicular to the line connecting the two terminals of the pacing electrode with the proximal microelectrode placed ~1 mm from the pacing electrode (Fig. 1A). This spacing is 3-6 times longer than the 0.3 mm passive length constant of bullfrog ventricular tissue [38]. Voltage signals acquired at 1 kHz were low-pass filtered (1 kHz 3-dB bandwidth) with analog circuitry and stored on a computer for later processing.

Restitution Portrait

Action potential durations and diastolic intervals were found using a threshold of 70% of the action potential amplitude. Restitution portraits were generated for each electrode by plotting

APD versus previous DI as shown in Fig. 2. RPs contain the transient response (small dots), which ends with the steady state responses (diamonds); the responses to S2 paces at BCL+50 ms ('+') and at BCL-50 ms ('x'); and the response to recovery paces (filled circles). These responses form the following RCs: the DRC (dashed line) that runs through all the steady state responses; segments of SRCs (grey lines) that are determined by the S2 paces and the steady state response at each BCL; and segments of BRCs (black lines) that are determined by the two sets of five recovery paces and the steady-state response at each BCL.

Each RP provides four measures that are used to characterize the tissue at each BCL:

1. The steady-state APD, computed as an average of the five steady-state responses (diamonds in Fig. 2).
2. Slope of the SRC (S_{SRC}). To compute S_{SRC} , the S2 ('+' and 'x') and steady state (diamonds) paces forming a segment of SRC are fit to a straight line using least-square regression (grey lines in Fig. 2). The slope of this line determines S_{SRC} at the steady-state of the BCL under consideration.
3. Slope of the BRC (S_{BRC}) is computed like S_{SRC} but using the recovery (circles) and steady state (diamonds) paces forming a segment of BRC (black lines in Fig. 2).
4. Slope of the DRC (S_{DRC}). Steady-state APDs for all BCL are fit to an exponential function,

$$APD = A - B \exp(-DI/\tau), \quad (1)$$

where A , B , and τ are parameters. S_{DRC} is computed by evaluating the derivative of Eq. 1 at the BCL under consideration.

These four characteristics were determined for each electrode at every BCL in the downsweep.

In six frogs, we also determined the time constant of the transient response following a downstep

in BCL. However, spatial differences in the time constant were not found to be statistically significant, and this characteristic is not included in further analysis.

Spatial Differences

The four characteristics of the RP were used to investigate spatial differences in restitution. Differences in APD were computed by subtracting steady-state APD at the distal electrode from the one at the proximal electrode: $\Delta APD = APD^{prox} - APD^{dist}$. Differences in slopes, ΔS_{SRC} , ΔS_{BRC} , ΔS_{DRC} were computed likewise. All computations were performed in Matlab 6.5.1 (The MathWorks, Natick, MA).

Spatial differences were analyzed as a function of the pacing rate. To properly compare results from different trials, BCL was shifted by subtracting the ‘transition’ BCL at which either 2:2 or 2:1 behavior was observed (BCL_t); that is, $BCL_s = BCL - BCL_t$. All shifted BCLs were divided into three pacing groups: slow pacing ($BCL_s > 400$ ms), medium pacing ($200 \text{ ms} < BCL_s \leq 400$ ms), and fast pacing ($BCL_s \leq 200$ ms). ALT and noALT trials were analyzed separately.

To assess whether the measurements made at the two different spatial locations differ, our study uses the statistical method proposed by Altman and Bland [39]. This method determines whether there is a spatial difference in, say, APD by constructing histograms of ΔAPD and testing whether the mean of the ΔAPD histogram (denoted by $\overline{\Delta APD}$) is zero using a t-test [40]. A $\overline{\Delta APD}$ significantly different from zero suggests that APDs differ at the two spatial locations. We then compare ΔAPD histograms for ALT and noALT trials. In this case, the t-test compares the means of the histograms to determine if there is a significant difference in $\overline{\Delta APD}$ between

ALT and noALT trials. The same method is used to investigate spatial differences in SRC, BRC, and DRC slopes. The analysis is performed separately for the three pacing groups, slow, medium, and fast. Statistical tests were performed in Excel (Microsoft, Redmond, WA). In all t-tests, p value less than 0.05 was considered significant.

Slope Criteria for the Onset of Alternans

For completeness, we analyzed our data to determine whether the traditional slope criteria can distinguish between ALT and noALT trials at either proximal or distal electrode. We extracted from the data the mean values of S_{SRC} , S_{BRC} , and S_{DRC} at all $BCL_S > 0$. In addition to examining individual slopes, we also evaluated a ‘memory’ criterion that involves a combination of SRC, BRC, and DRC slopes and accounts for one possible way that short-term memory affects rhythm stability [18]. The ‘memory’ criterion is given by the following combination of slopes:

$$S_{mem} = \left| 1 - S_{SRC} - \frac{S_{SRC}}{S_{DRC}} + \frac{S_{SRC}(1 - S_{DRC})}{S_{DRC} - S_{SRC}} \left(1 - S_{SRC} - \frac{S_{SRC}}{S_{DRC}} + S_{BRC} \right) \right|. \quad (2)$$

These four quantities were evaluated separately for data from the proximal and distal electrode.

For any quantity S , the criterion for alternans is considered satisfied when: 1) S is less than one for all $BCL_S > 0$ for the ALT and noALT trials; 2) S increases as BCL_S decreases for the ALT and noALT trials; and 3) S approaches one as BCL_S approaches zero for the ALT trials but not for the noALT trials.

Results

Restitution Portraits

Figure 2 shows the restitution portraits generated from simultaneous measurements with two microelectrodes proximal and distal to the stimulus site for one of the ALT trials. These restitution portraits have the same qualitative features as those measured previously in bullfrog [19], rabbit, and guinea pig [13]. RPs from both locations are qualitatively similar, although there is evidence of quantitative differences between them. In particular, at all BCLs, there is a large difference in APDs between the two locations, about 30-50 ms, and the DRC for the proximal electrode is visibly steeper at short BCLs. Overall, 19 pairs of simultaneously-measured RPs are constructed and the differences between them are analyzed below.

Steady State APD

At all pacing rates, the spatial difference of APD is positive (Figs. 3A and 4; Table 1), indicating that the APD decreases as the wave propagates away from the stimulus. $\overline{\Delta APD}$ is significantly different from zero at all pacing rates and for both ALT and noALT trials (p values given in the panels of Fig. 4). Furthermore, $\overline{\Delta APD}$ s in ALT trials are significantly larger than in noALT trials in all pacing groups (p values given on the right in Fig. 4).

SIS2 Restitution Curve

The mean SRC slope difference ($\overline{\Delta S_{SRC}}$) is positive in ALT trials and negative in noALT trials (Figs. 3B and 5; Table 1). $\overline{\Delta S_{SRC}}$ moves toward zero as the pacing rate increases, still remaining positive in ALT trials and negative in noALT trials. P values given in the panels of Fig. 5 show that $\overline{\Delta S_{SRC}}$ is significantly different from zero for ALT trials; for noALT trials, $\overline{\Delta S_{SRC}}$ is

significantly different from zero for slow and medium pacing but not for fast pacing. Note that $\overline{\Delta S}_{SRC}$ in ALT and noALT trials differs significantly at all pacing rates (p values given on the right in Fig. 5).

Constant-BCL Restitution Curve

The mean BRC slope difference ($\overline{\Delta S}_{BRC}$) is generally positive in ALT trials and negative in noALT trials (Fig. 3C and 6; Table 1). $\overline{\Delta S}_{BRC}$ moves toward zero as BCL decreases, still remaining positive in ALT trials and negative in noALT trials. This is similar to what is seen with $\overline{\Delta S}_{SRC}$, but the effect is smaller in $\overline{\Delta S}_{BRC}$. P values given in the panels of Fig. 6 show that $\overline{\Delta S}_{BRC}$ is significantly different from zero at slow and medium pacing for ALT trials; for noALT trials, $\overline{\Delta S}_{BRC}$ is not significantly different from zero only at any pacing rate. Note that $\overline{\Delta S}_{BRC}$ values in ALT and noALT trials differ significantly at slow and medium pacing rates, but not at fast pacing (p values given on the right in Fig. 6).

Dynamic Restitution Curve

The mean DRC slope difference ($\overline{\Delta S}_{DRC}$) is mostly positive in both ALT trials and noALT trials (Fig. 3D and 7; Table 1). $\overline{\Delta S}_{DRC}$ increases as BCL decreases, particularly in ALT trials (Fig. 3D). P values given in the panels of Fig. 7 show that $\overline{\Delta S}_{DRC}$ is significantly different from zero only at fast pacing rates for ALT trials; for noALT trials, $\overline{\Delta S}_{DRC}$ is significantly different from zero only at slow pacing. Note that $\overline{\Delta S}_{DRC}$ values in ALT and noALT trials differ significantly at slow and fast pacing, but not at medium pacing (p values given on the right in Fig. 7).

Slope Criteria for the Onset of Alternans

We have found that none of the slopes of individual restitution curves (S_{SRC} , S_{BRC} , S_{DRC}) or the ‘memory’ criterion, evaluated from S_{mem} , correlate with alternans. Specifically, the SRC and BRC are generally very shallow and their slopes remain well below one (less than or of the order of 0.3) for all experiments. In contrast, DRC becomes quite steep at small BCL with slopes well above one seen in 59% of all measurements that show stable 1:1 responses for both ALT and noALT trials. The ‘memory’ criterion fails because S_{mem} decreases as BCL_S decreases and it is much larger than one for slow pacing (large BCL_S).

Discussion

Spatial Differences in Restitution Properties

Our study found spatial differences in several characteristics of restitution. In particular, the steady state APD and the slope of the SRC showed significant spatial differences at all pacing rates in both ALT and noALT trials with a single exception of ΔS_{SRC} of noALT trials at fast pacing. APD showed a consistent positive difference, suggesting that the APD was larger close to the stimulus. SRC slope showed a mean positive difference in ALT trials and a mean negative difference in noALT trials, although this trend did not hold for every trial. Results for ΔS_{BRC} and ΔS_{DRC} were mixed with significant spatial differences detected only at some pacing rates. Specifically, the slope of the BRC showed significant spatial differences at medium and slow pacing in ALT trials, but not at any pacing rate in noALT trials. The slope of the DRC showed significant spatial differences only at fast pacing in ALT trials and slow pacing in noALT trials.

We believe that the spatial differences we observed are dynamically induced because of our choice of experimental substrate. Amphibian hearts do not have Purkinje fibers [41] or vasculature [35-37]. There is also no evidence of different types of cells in the ventricle nor any evidence of fiber structure on a macroscopic scale [35,36]. The homogeneous nature of bullfrog ventricular tissue suggests that any observed spatial differences in restitution properties are not simply reflecting underlying tissue inhomogeneity, but are instead created dynamically during pacing of the tissue.

Some spatial differences reported here have been seen in experiments using mammalian cardiac tissue. Large APD gradients seem to be a common characteristic of all cardiac tissue, having been observed in guinea pig [4,22,25,26,29] and pig [23,27] hearts. However, the APD gradients observed in those experiments were smaller than those observed in our experiments. In simulations of homogeneous myocardium [42,43], APD gradients were also larger than in experiments, suggesting that tissue heterogeneity may decrease spatial variation of APD. Choi and Salama [25] observed shorter APDs at the apex and longer APDs at the base, further suggesting that spatial tissue heterogeneity may alter APD gradients. Gradients in the slope of the SRC have also been observed in previous studies [28-31], however these measurements were made in humans [28] and guinea pigs [29-31], where the gradients may have been caused by fiber structure or other heterogeneities. In fact, Akar *et al.* note that the SRC slope gradient is parallel to the cardiac fibers [29]. Finally, our observation that the DRC slope differences were primarily positive differs from the findings of Qin *et al.* [27], who observed no consistent gradient in DRC. These findings suggest that tissue heterogeneity in mammals may mask dynamically-induced differences in slope of the DRC.

Predicting the Tissue's Propensity to Alternans

The restitution hypothesis [6], in which the stability is determined by the slope of the restitution curve, failed in all trials, regardless of the slope tested or the measurement location. The slopes of the SRC and BRC remained well below one in ALT trials, even near the onset of alternans. In fact, the slopes of the SRC and BRC were slightly larger in noALT trials just before the transition to a 2:2 response. The slopes of the DRC were quite steep even during stable 1:1 response. This result has also been observed in other experiments [12,14,19] and in simulations [12,15,19,21].

The failure of the restitution hypothesis is not surprising since our experiments show rate-dependence and short-term memory (Fig. 2). However, even a stability criterion that takes into account these effects in one particular form (Eq. 2) does not accurately predict the tissue's propensity to alternans. In our experiments, S_{mem} was well above one during stable 1:1 response in ALT and noALT trials. The failure of all slope-based criteria suggests the models of local APD dynamics, from which these criteria are derived, do not completely capture the factors responsible for the loss of stability in spatially-extended cardiac tissue.

Some of the factors not accounted for by the slope criteria include border-collision bifurcations, intracellular calcium, and spatial interactions. Border-collision bifurcation occurs in systems whose dynamics is described by a piece-wise smooth function [44]; a bifurcation occurs when the parameter crosses the border between two smooth pieces of the function. Recent research has suggested that the transition to alternans in paced cardiac tissue may be mediated by a

bifurcation that has border-collision characteristics [45]. If this is the case, it may not be possible to predict the onset of alternans by examining the 1:1 response. It would explain why the slope-based criteria that are derived from models with smooth dynamics fail to predict the onset of alternans.

Another possibility is that APD alternans are driven by calcium alternans [46-51]. In mammals, the stability of calcium transients is thought to be determined by the feedback gain of calcium release from the sarcoplasmic reticulum [50]. This possibility shifts the problem of predicting the onset of APD alternans to one of predicting the onset of calcium alternans. A stability criterion for calcium alternans [50] involves measures of calcium changes in the cytosol and sarcoplasmic reticulum, but the measurement of subcellular calcium has only recently been achieved during repeatable fast pacing [51]. However, calcium cycling in frog myocardium relies much less on the sarcoplasmic reticulum [52], so calcium alternans may not drive APD alternans or they may do so by a different mechanism in our experiment.

Finally, all stability criteria used to date were derived from models of single cardiac cell dynamics. Our experiments, and most other cardiac experiments related to rhythm stability, are performed in spatially extended pieces of tissue where neighboring cells are coupled through gap junctions. Experiments and simulations show that cell-to-cell coupling strongly affects cardiac rhythm dynamics: it decreases spatial heterogeneity [42,53] and changes the pacing rate at which alternans are observed [42,54-56].

The results presented here indicate that spatial interactions may also play a critical role in the stability of the 1:1 response. We found that spatial differences in APD and SRC slope are predictive of alternans at all pacing rates. ALT trials had greater $\overline{\Delta APD}$ than noALT trials and $\overline{\Delta S_{SRC}}$ was positive in ALT trials but negative in noALT trials. The slope of the DRC is predictive of alternans at slow and fast pacing, with $\overline{\Delta S_{DRC}}$ becoming positive and growing in magnitude at fast pacing rates in ALT trials. Finally, the slope of the BRC is predictive of alternans at slow and medium pacing rates. Most importantly, all four characteristics are predictive of alternans at *slow* pacing rates, which may have a clinical advantage, as discussed below.

Study Limitations and Future Work

Our findings are purely empirical. In contrast to the traditional slope criterion, we do not have an underlying model and cannot offer a theoretical explanation of why the spatial differences should be predictive of alternans. Likewise, without an underlying model, we cannot propose a specific quantitative threshold for alternans. Nevertheless, the results are compelling enough to stimulate future work on developing the theory linking spatial differences in restitution properties and tissue's propensity to alternans.

The first step in building such a theory is to obtain a better picture of the observed spatial differences. Use of only two microelectrodes can tell us whether spatial differences exist, but it does not give us any details of the spatial distribution. The repeatability of our results over seven animals suggests that these differences result from consistent spatial gradients of restitution properties, rather than from random patterns. Nevertheless, there is a need for a follow-up study

that would use potentiometric dyes and an optical imaging system to provide information about the spatial distribution of multiple restitution properties throughout the tissue as a function of pacing rate.

The results presented in this study may be specific to bullfrog tissue, which is relatively uniform, both structurally and electrophysiologically [35-37, 41]. To be clinically useful, our results will need to be confirmed in mammalian tissue. The main question is whether the more complex tissue structure in mammals may alter dynamically-induced spatial gradients of restitution. As discussed above, spatial gradients of APD, S_{SRC} , and S_{DRC} have also been observed in mammalian tissue [4,23-31], suggesting that some of our observations may hold in larger, heterogeneous hearts. A simulation study has shown that dynamically-induced spatial gradients in APD appear on the surface only in small and relatively uniform hearts [43]; in larger and more heterogeneous hearts, these gradients tend to appear transmurally. It is not known whether these transmural gradients show the same correlation to the onset of alternans as observed in our experiments on bullfrogs. Studies of rabbit hearts are currently underway in our laboratory to confirm the findings presented here.

Clinical Implications

Our results, if they are confirmed in mammalian tissue, may have important implications for the diagnosis and treatment of patients with arrhythmias. Diagnostic procedures, either currently in use or proposed based on theory and experiments, analyze spatially-averaged temporal response patterns (microvolt T-wave alternans in the ECG [57]) or temporal response patterns at a single location (steepness of the restitution curve [32] or increase in gain during alternate pacing [58]).

Our study suggests an entirely new approach, based on spatial differences in restitution properties. Likewise, while some current pharmacological treatments aim to prevent arrhythmias by decreasing steepness of the restitution curve [59], our results may open the possibility of alternative pharmacological treatments that target spatial gradients of restitution properties. Thus, our findings may lead to entirely new methods of determining a patient's vulnerability to arrhythmias and to the design of novel antiarrhythmic medications.

The proposed spatial approach may have significant advantages. Existing interventional diagnostic methods include pacing at fast rates or delivering multiple shortly-coupled extrastimuli in order to induce an episode of an arrhythmia. Our results show that spatial differences are predictive of alternans at cycle lengths much longer than the cycle length of the transition to alternans. This suggests that the new procedure may be safer and better tolerated by patients. Additionally, it can be implemented using existing technology because diagnostic catheters already have multiple recording sites.

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Table 1: The mean and standard error of differences in APD and in slopes of SRC, BRC, and DRC.

	ALT trials			noALT trials		
	Slow Pacing	Medium Pacing	Fast Pacing	Slow Pacing	Medium Pacing	Fast Pacing
$\overline{\Delta APD}$	63±8 ms	83±8 ms	71±8 ms	38±4 ms	32±5 ms	25±4 ms
$\overline{\Delta S}_{SRC}$	0.4±0.1	0.35±0.07	0.18±0.06	-0.22±0.06	-0.16±0.06	-0.08±0.03
$\overline{\Delta S}_{BRC}$	0.4±0.1	0.3±0.07	0.04±0.09	-0.44±0.05	-0.07±0.07	-0.04±0.06
$\overline{\Delta S}_{DRC}$	-0.13±0.07	0.2±0.1	1.2±0.1	0.22±0.06	0.02±0.08	0.17±0.08

Figure Captions

Figure 1: (A) Sketch of a bullfrog ventricular preparation. Two microelectrodes are placed 1-2 mm apart with the proximal one placed ~ 1 mm from the bipolar pacing electrode. (B) A sequence of the perturbed downsweep pacing protocol for one BCL. The tissue is paced at a constant BCL for 60 s (transient response, small dots). Additional 5 paces at steady state are applied (diamonds) followed by an S2 pace at BCL+50 ms ('+'), 5 recovery paces at the original BCL (filled circles), an S2 pace at BCL-50 ms ('x'), and 5 more recovery paces (filled circles). The entire sequence is repeated at progressively shorter BCLs until the myocardium transitions to a 2:1 or 2:2 stimulus:response pattern. Downstep in BCL, Δ , is 50 or 100 ms.

Figure 2: Restitution portraits collected simultaneously from the electrode proximal (A) and distal (B) to the pacing site. The restitution portraits contain all the responses of the perturbed downsweep protocol of Fig. 1B: the transient response (small dots), the steady state responses (diamonds), the S2 pace at BCL+50 ms ('+'), the S2 pace at BCL-50 ms ('x') and the recovery paces (circles). The DRC (dashed line) is the curve that connects all steady state responses. At each BCL, segments of SRCs (grey lines) are determined by the S2 paces and the steady state response; segments of BRCs (black lines) are determined by the recovery paces and the steady-state response. For clarity, panels (A) and (B) show data for every second BCL collected in this trial.

Figure 3: Mean and standard error of APD and slope differences as a function of the shifted BCL. The vertical gray lines separate the BCLs assigned to the fast, medium and slow pacing rate groups.

Figure 4: Distributions of the difference in the steady-state APD ($\Delta APD = APD^{prox} - APD^{dist}$) at slow, medium, and rapid pacing in noALT and ALT trials. In each panel, the gray line indicates the mean of the distribution ($\overline{\Delta APD}$), while the black line indicates a ΔAPD of zero. P values in panels (A-F) indicate whether the distribution has a zero mean; p values to the right of the panels indicate whether the ALT and noALT trials have different distributions.

Figure 5: Distributions of the difference in the SRC slope ($\Delta S_{SRC} = S_{SRC}^{prox} - S_{SRC}^{dist}$) at slow, medium, and rapid pacing in noALT and ALT trials. In each panel, the gray line indicates the mean of the distribution ($\overline{\Delta S_{SRC}}$), while the black line indicates a ΔS_{SRC} of zero. P values in panels (A-F) indicate whether the distribution has a zero mean; p values to the right of the panels indicate whether the ALT and noALT trials have different distributions.

Figure 6: Distributions of the difference in the SRC slope ($\Delta S_{BRC} = S_{BRC}^{prox} - S_{BRC}^{dist}$) at slow, medium, and rapid pacing in noALT and ALT trials. In each panel, the gray line indicates the mean of the distribution ($\overline{\Delta S_{BRC}}$), while the black line indicates a ΔS_{BRC} of zero. P values in panels (A-F) indicate whether the distribution has a zero mean; p values to the right of the panels indicate whether the ALT and noALT trials have different distributions.

Figure 7: Distributions of the difference in the SRC slope ($\Delta S_{DRC} = S_{DRC}^{prox} - S_{DRC}^{dist}$) at slow, medium, and rapid pacing in noALT and ALT trials. In each panel, the gray line indicates the mean of the distribution ($\overline{\Delta S_{DRC}}$), while the black line indicates a ΔS_{DRC} of zero. P values in

panels (A-F) indicate whether the distribution has a zero mean; p values to the right of the panels indicate whether the ALT and noALT trials have different distributions.

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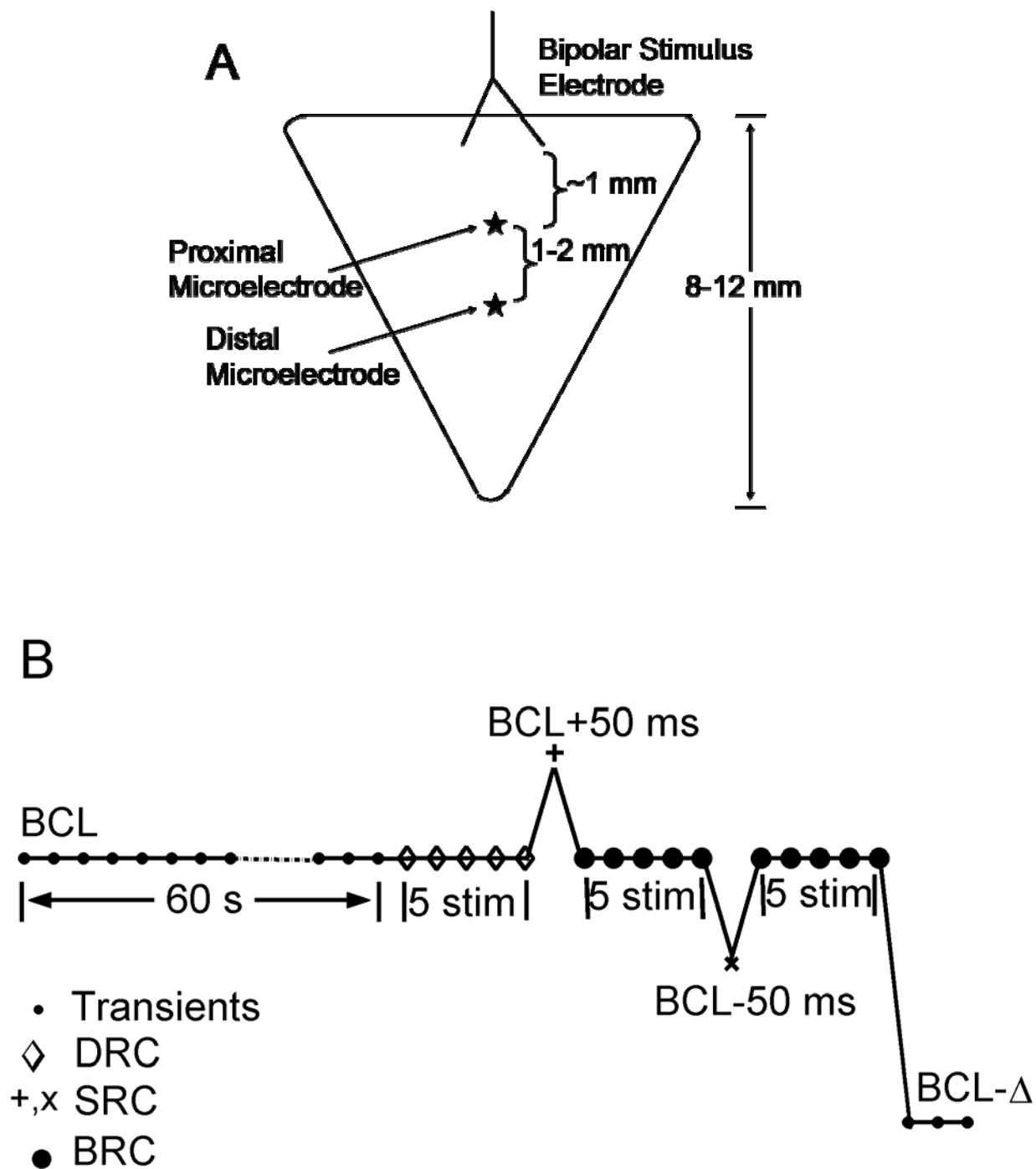


Figure 1

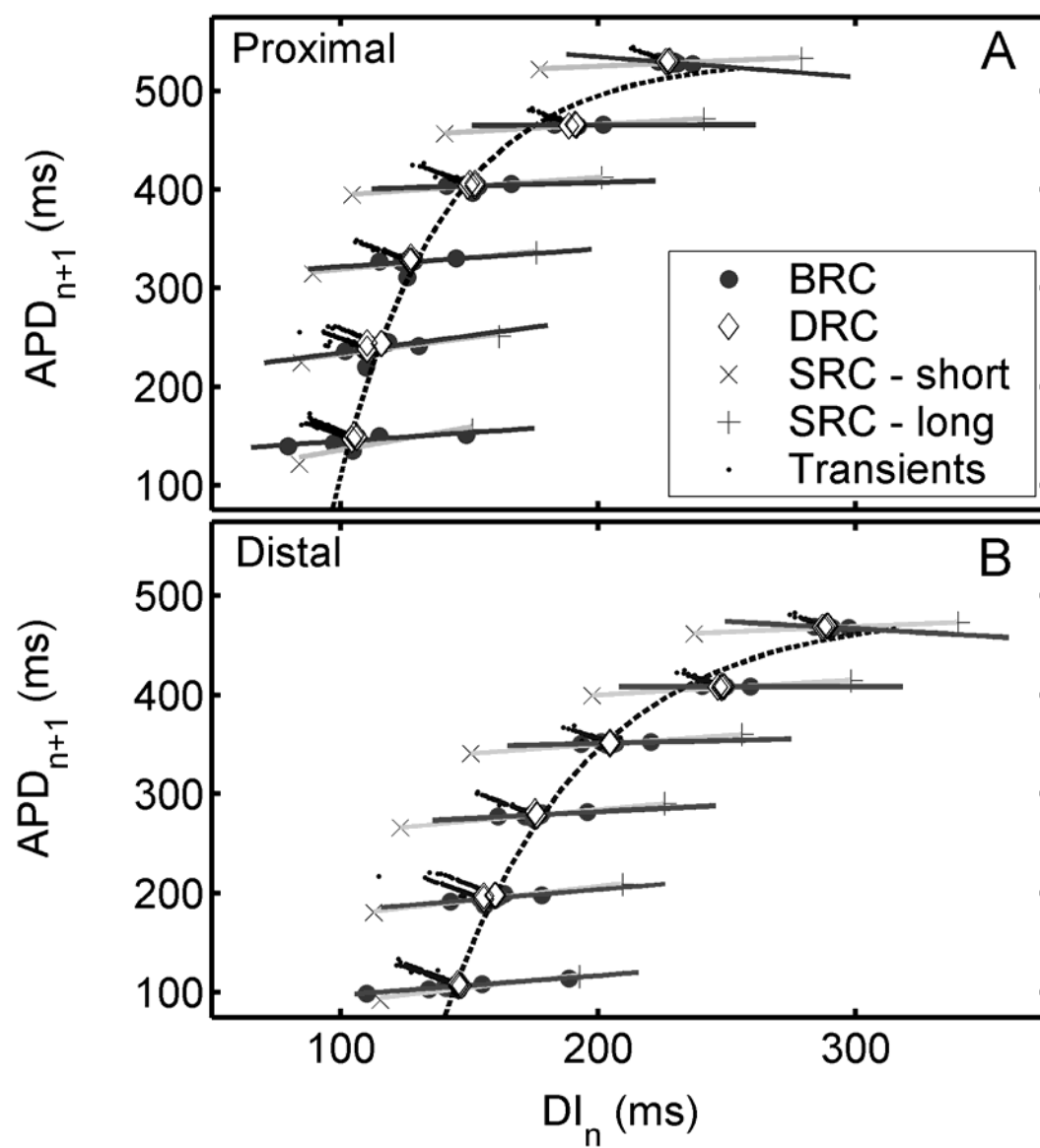


Figure 2

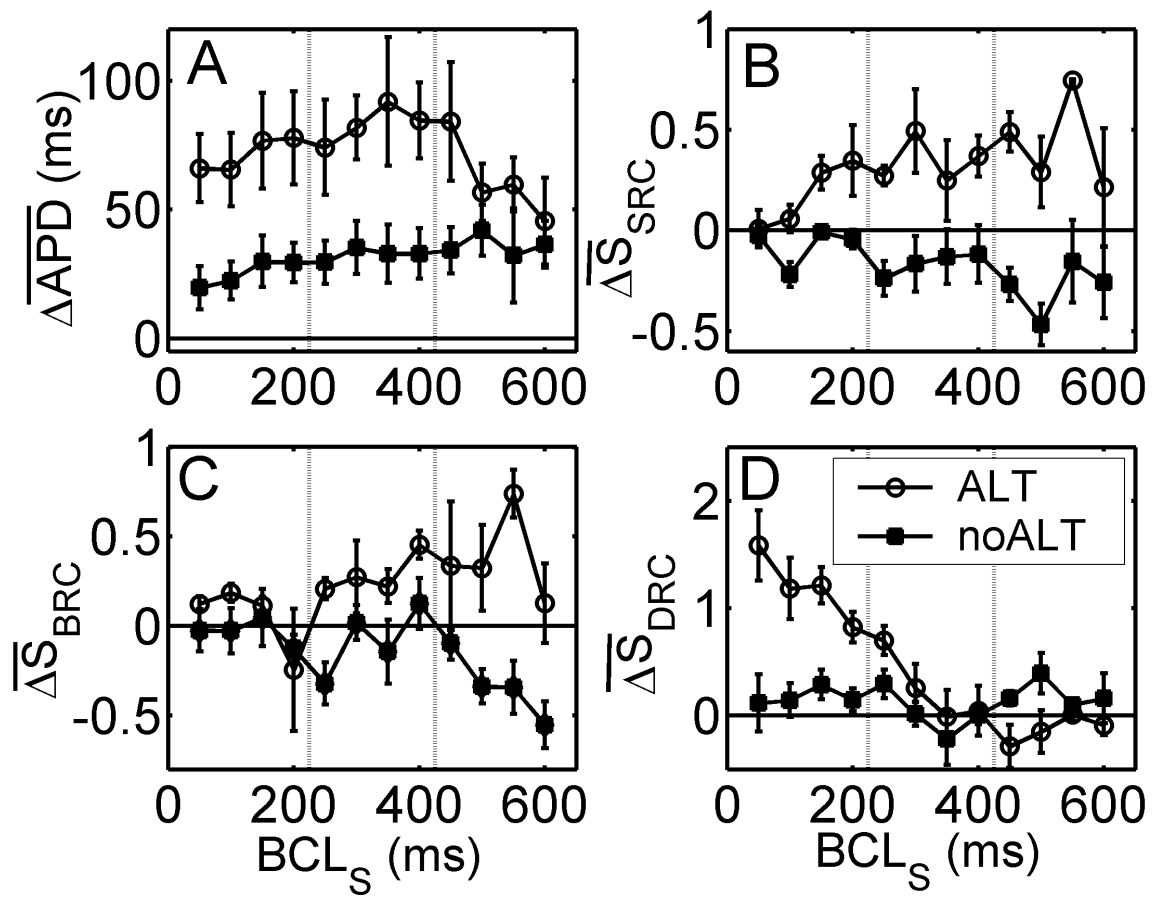


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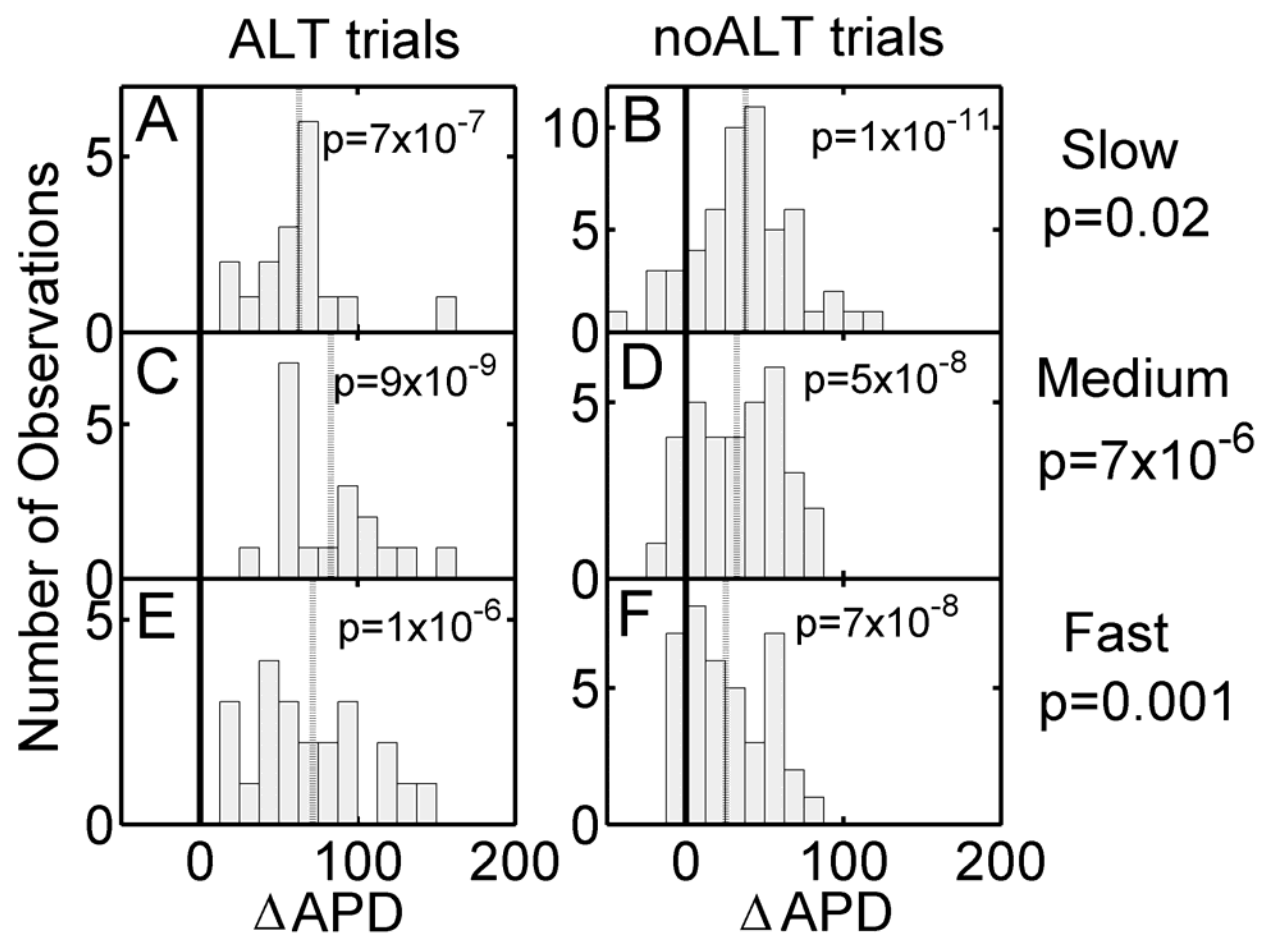


Figure 4

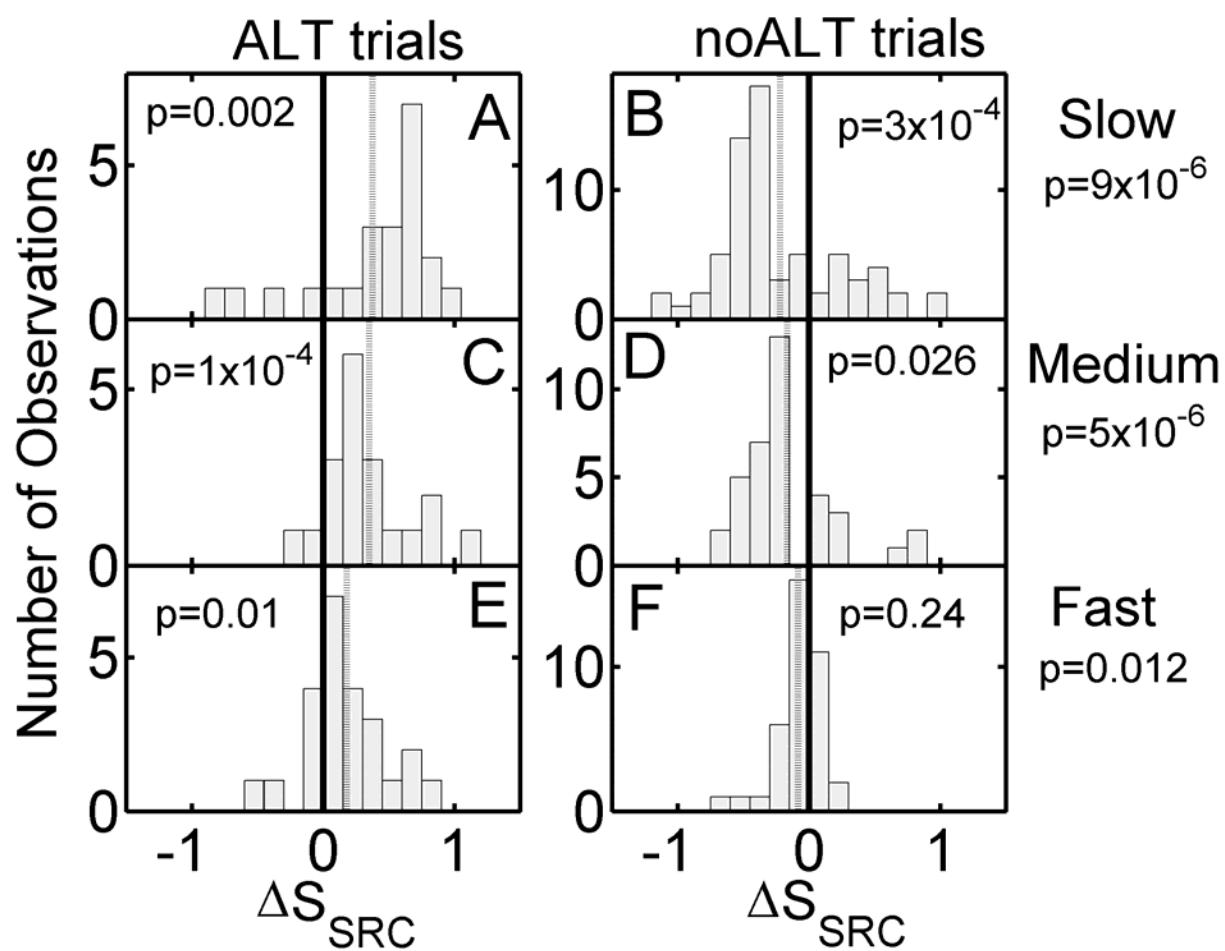


Figure 5

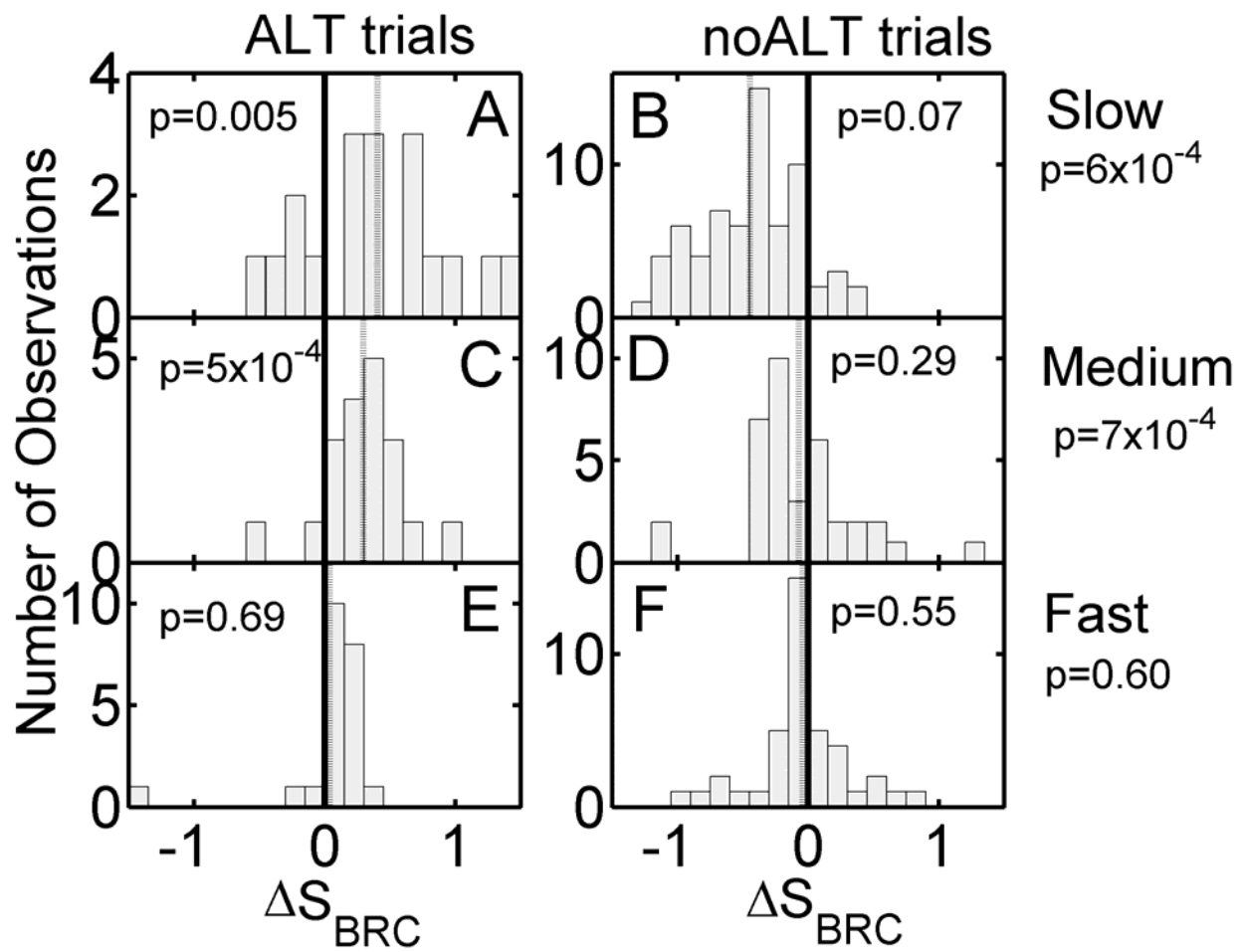


Figure 6

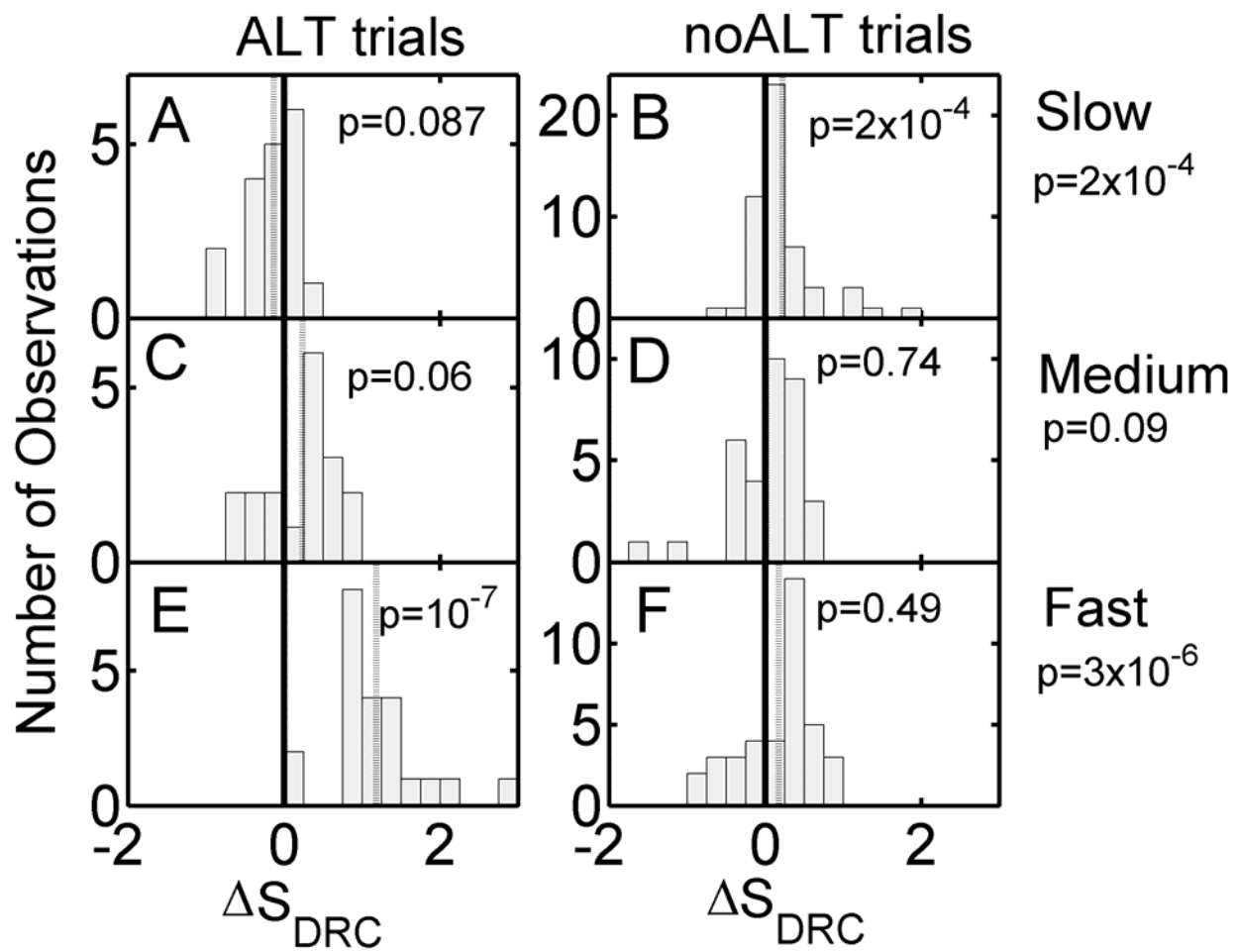


Figure 7